

Topic 8-2 Keras实现典型CNN

魏准
浙江大学，信息与电子工程学院
2026

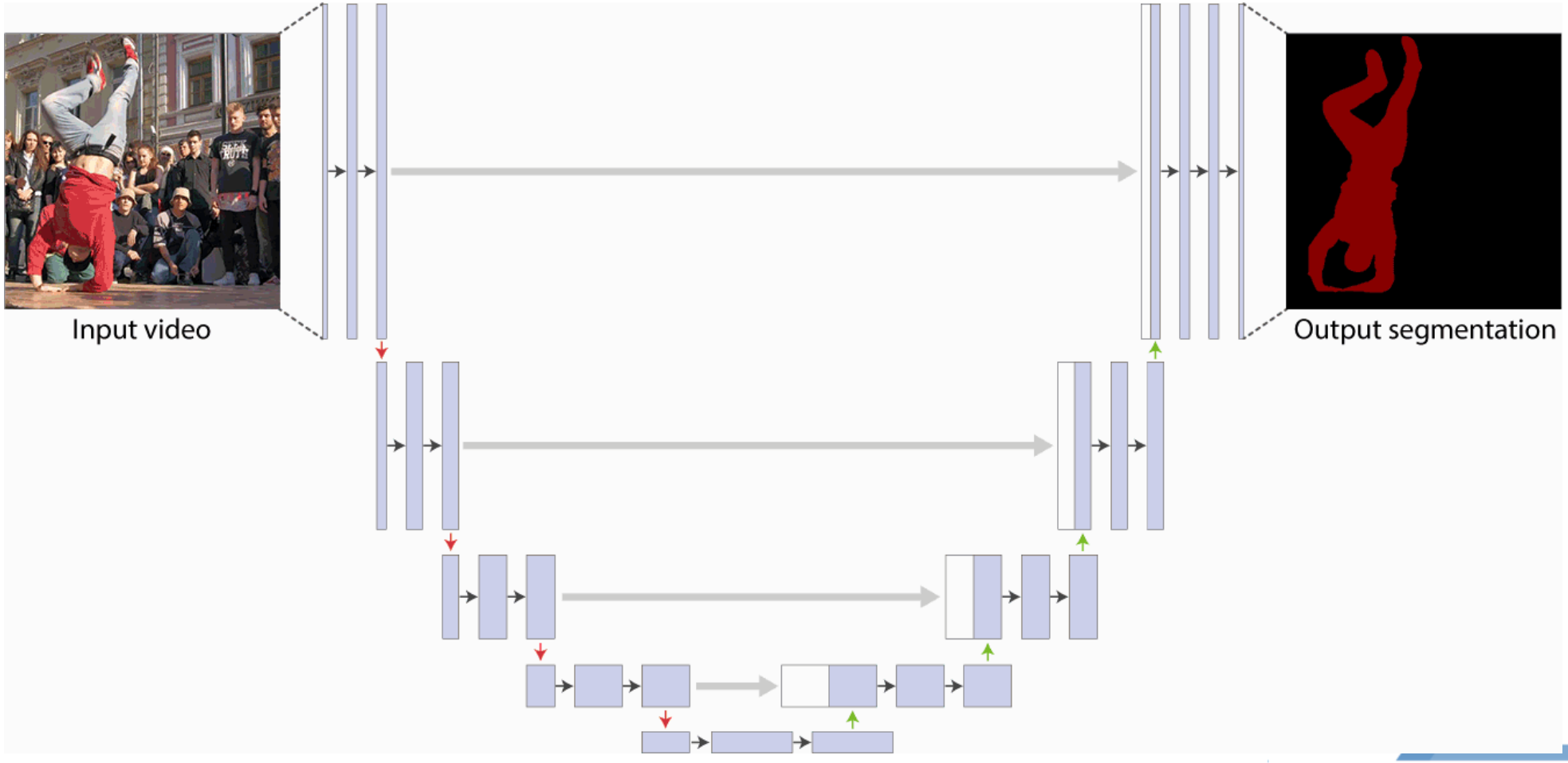
U-NET-Background

U-net: Convolutional networks for biomedical image segmentation

O Ronneberger, P Fischer, T Brox - … , Munich, Germany, October 5-9, 2015 …, 2015 - Springer

There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies on the strong use of data augmentation to use the available annotated samples more efficiently. The architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. We show that such a network can be trained end-to-end from very few images and outperforms the prior best method (a sliding …

☆ Save 99 Cite Cited by 62502 Related articles All 29 versions



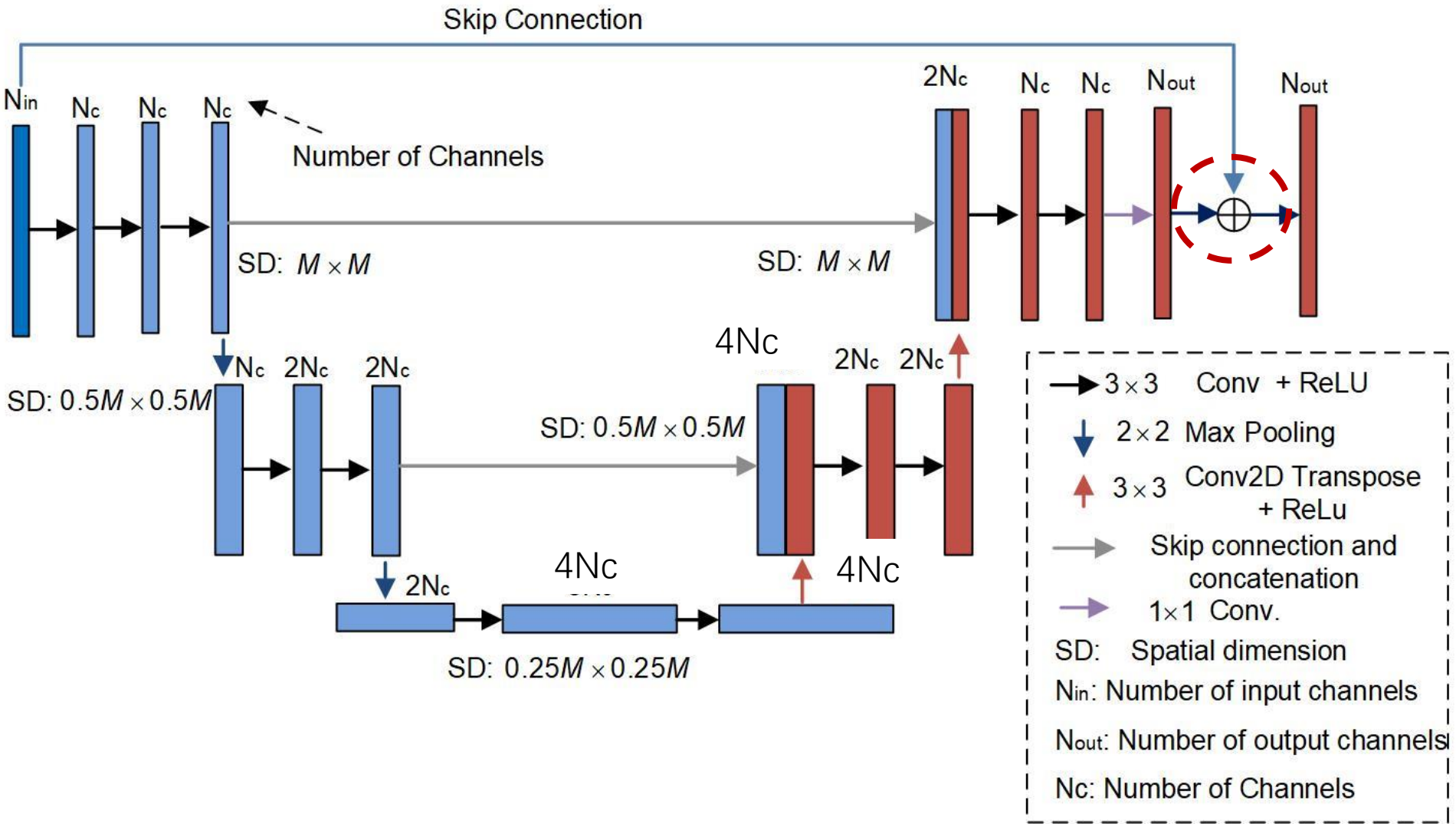
From: Image and Network Analysis Group

U-NET-Background



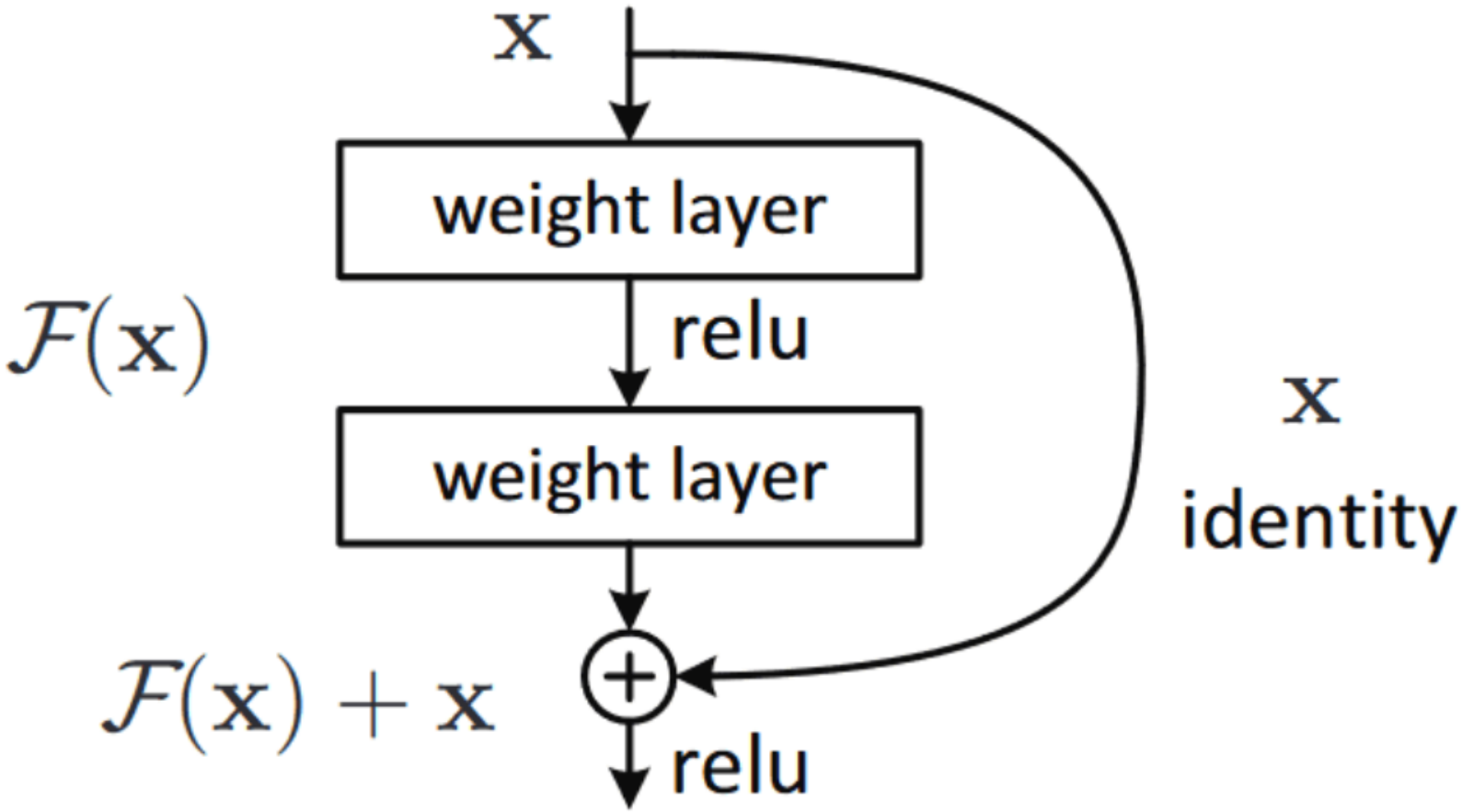
Image From http://casser.io/files/transfer_learning.html

U-NET



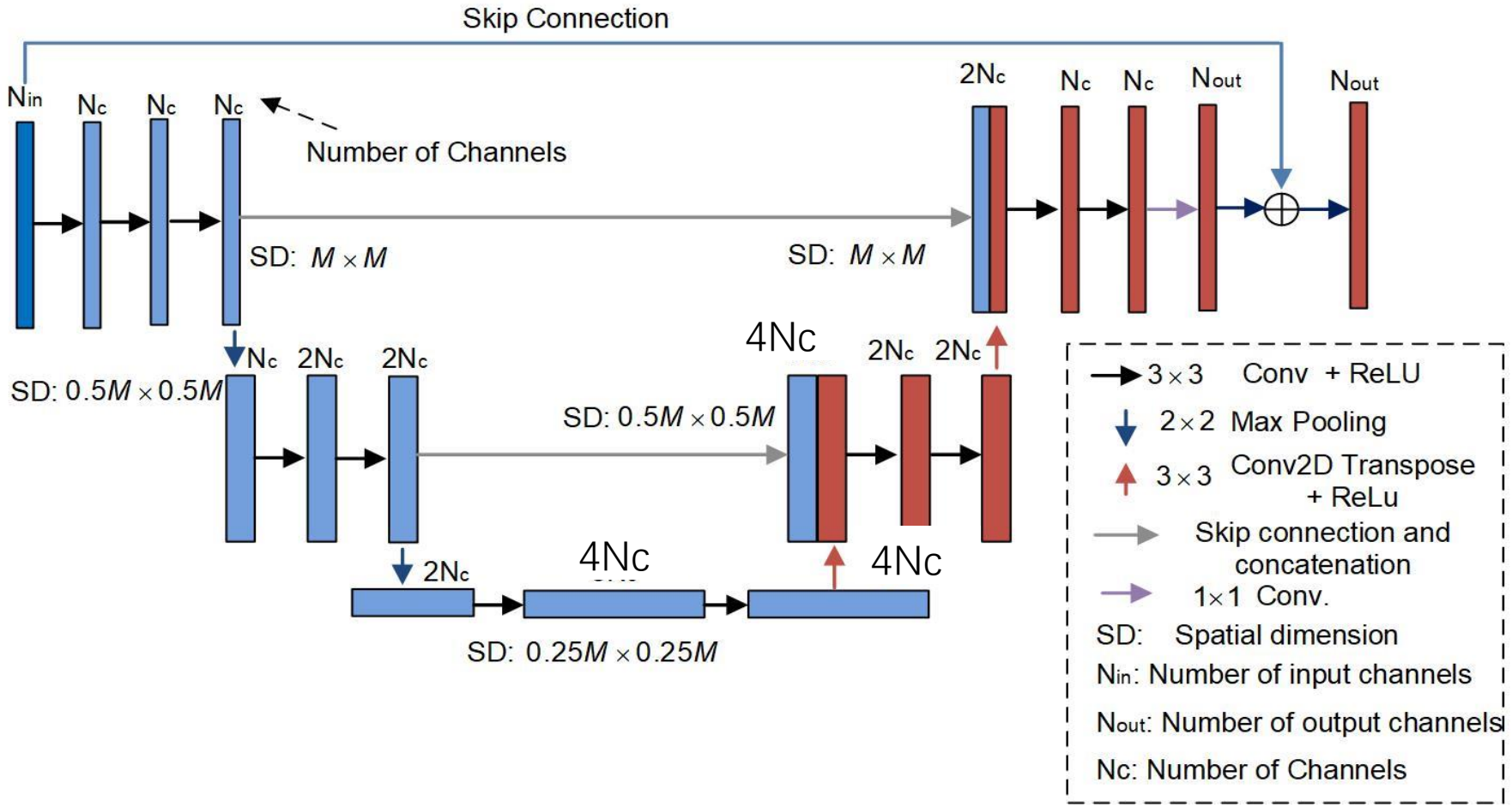
- Encoder-Decoder
- 编码器逐渐提取图
- 上、下文信息反复组合
- Skip Connection
- Concatenation
- 所需样本数量小

Skip-Connection(Residual Learning)



U-NET

- 综合信息
- 梯度更容易传到浅层
- 减轻任务目标




```
merge6 = concatenate([Conv3,up2], axis = -1)
added = Add()([conv12, inputs])
model = Model( inputs, added)
```

卷积基本特征

- 输入X： *W1×H1×D1*； （长宽通道）

filters： *F×F×D1×K;*

步长Stride: *S*;

补零数zero padding number: *P;*

- 输出： W2×H2×K;

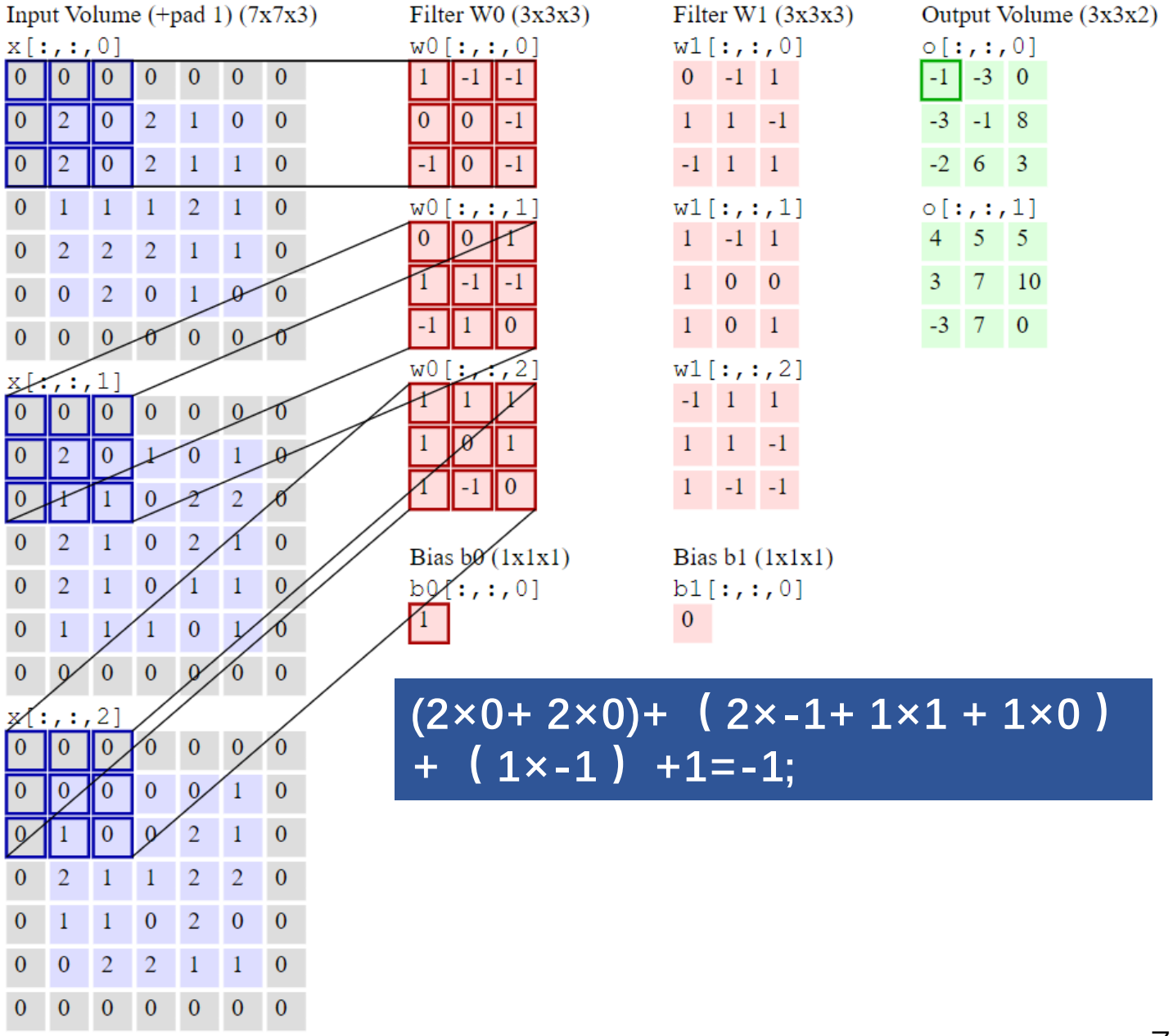
W2=(W1−F+2P)/S+1

=(5-3+2×1)/2+1=3;

H2=(H1−F+2P)/S+1

=(5-3+2×1)/2+1=3;

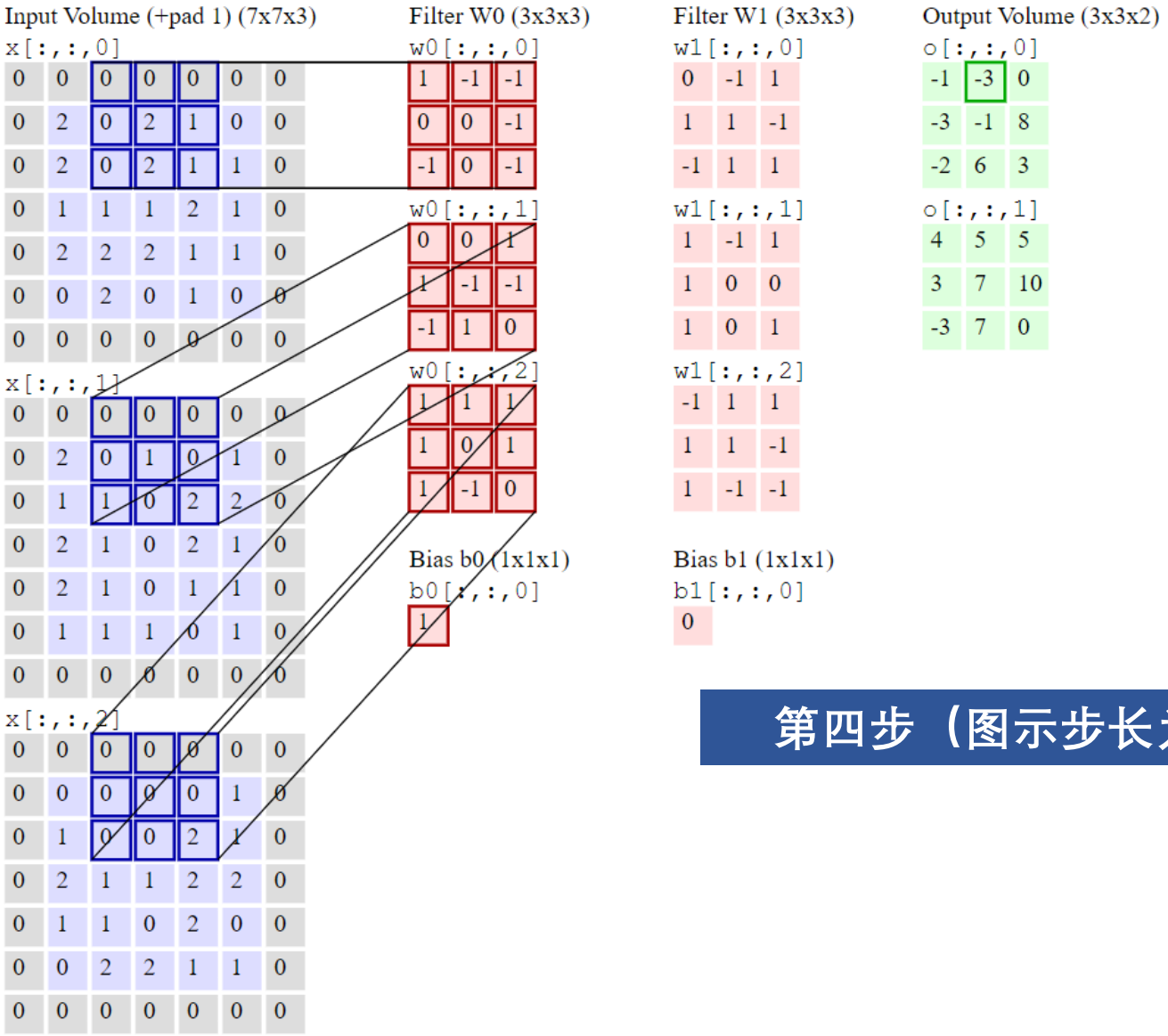
K=2;



(2×0+ 2×0)+（2×-1+ 1×1 + 1×0）+（1×-1）+1=-1;

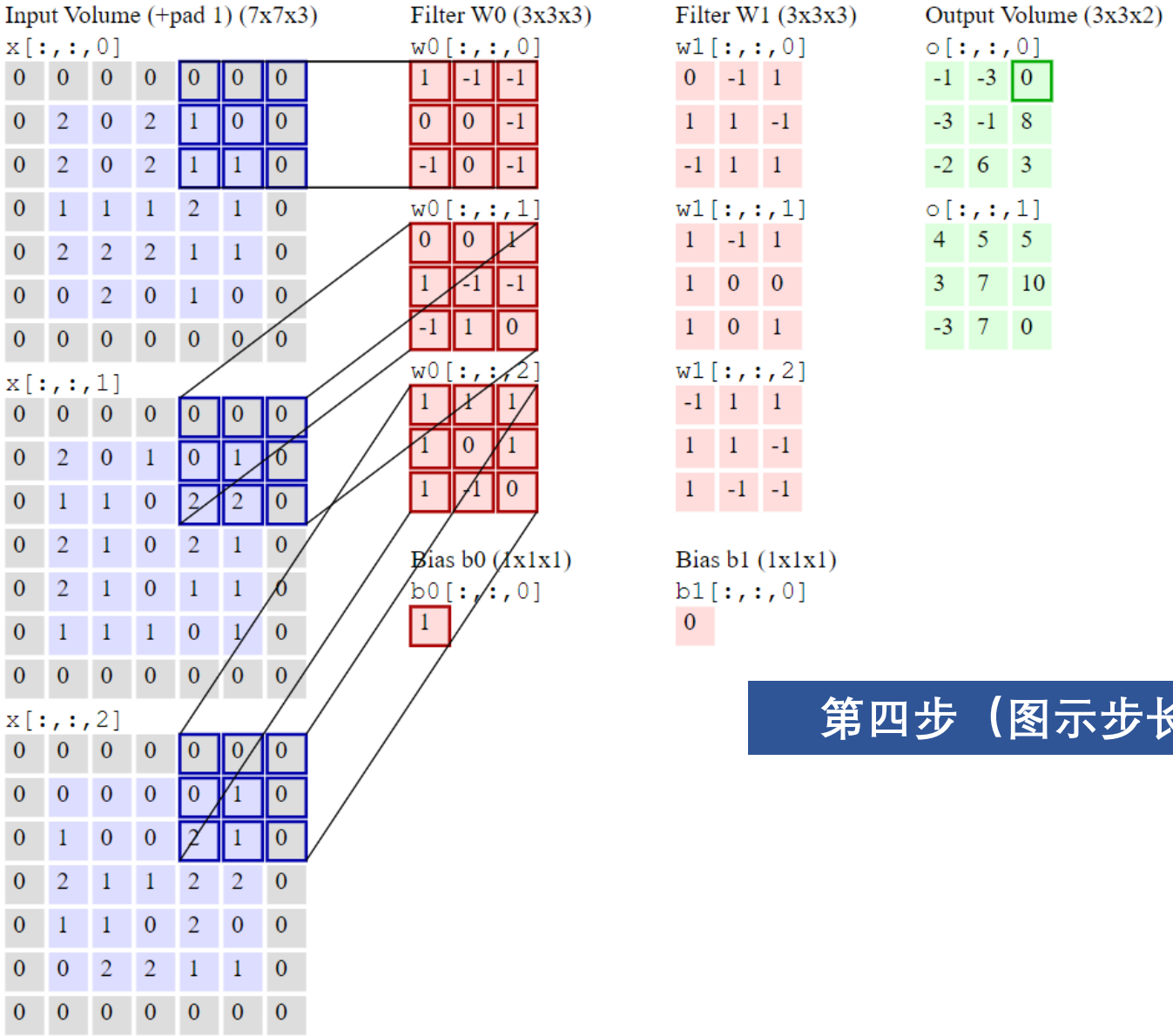
Source(CS231n: Deep Learning for Computer Vision)

基本特征



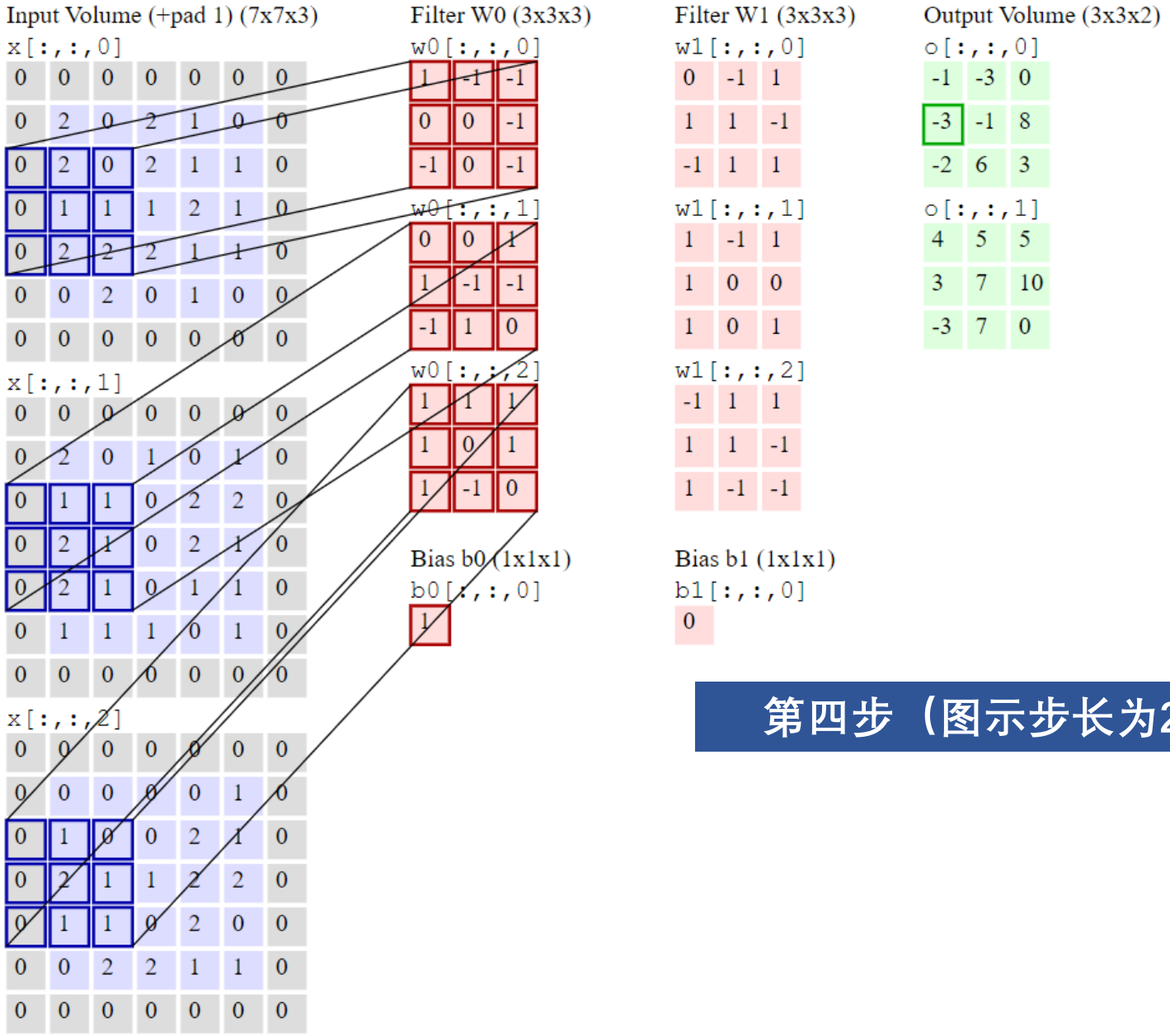
第四步（图示步长为2）！

基本特征



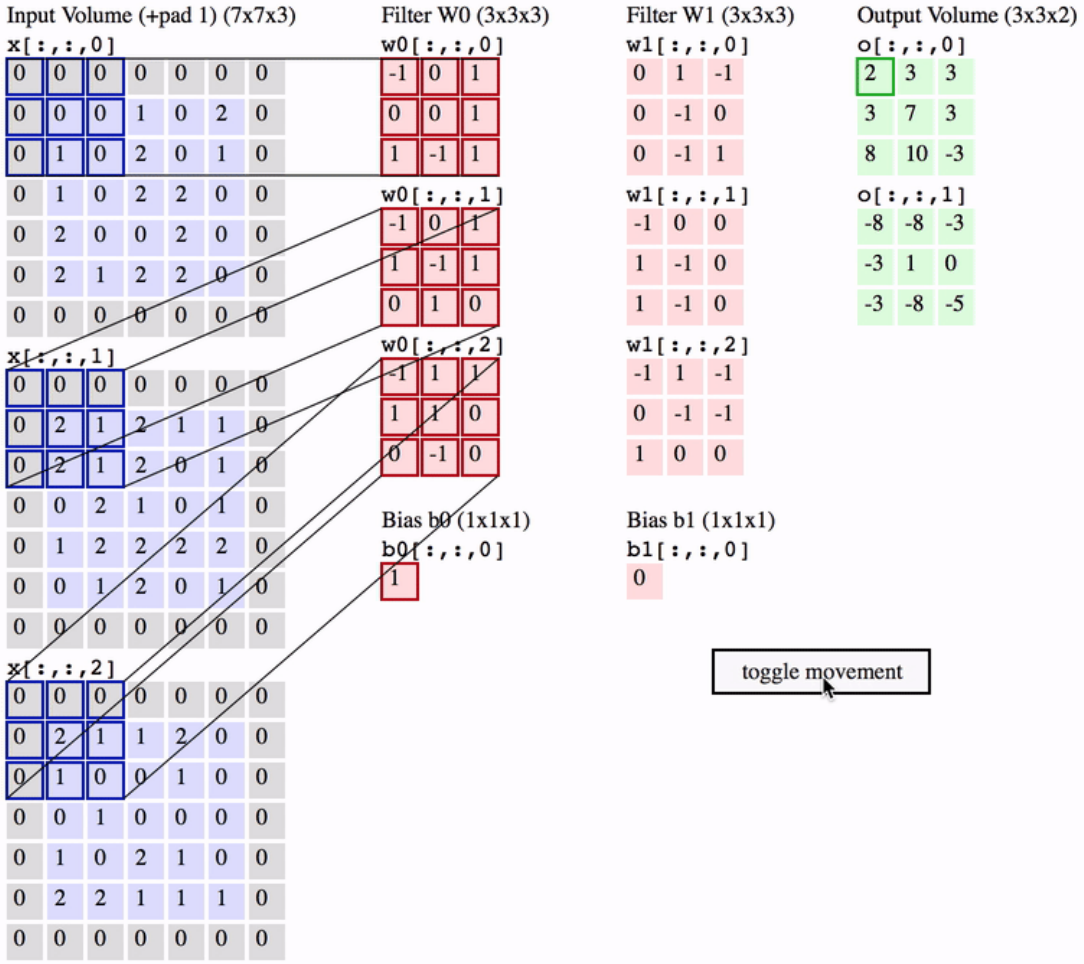
第四步（图示步长为2）！

基本特征



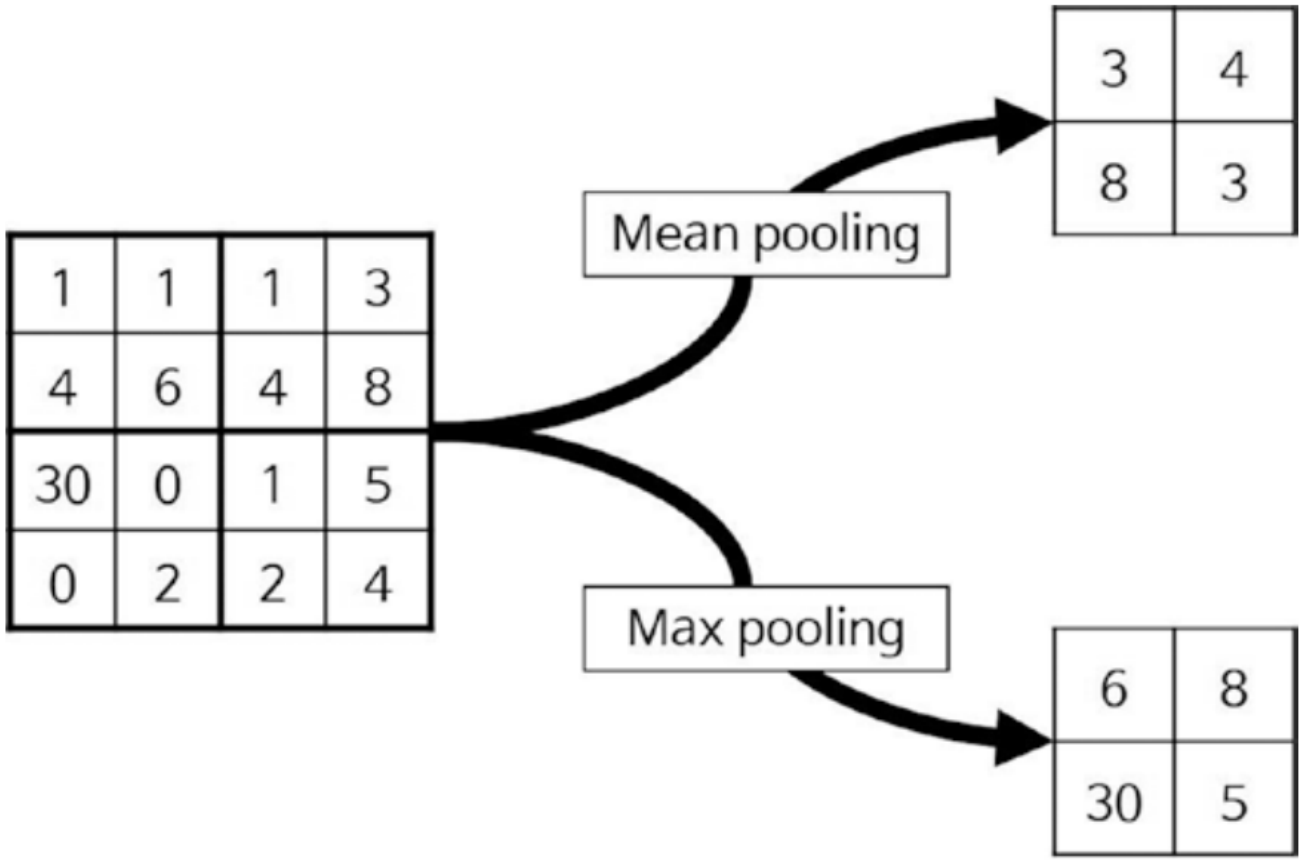
第四步（图示步长为2）！

基本特征




```
## 搭建模型
y = Conv2D(32, 3, padding='same', activation='relu',
kernel_initializer='he_normal’)(x)
```

池化层(Pooling Layer)

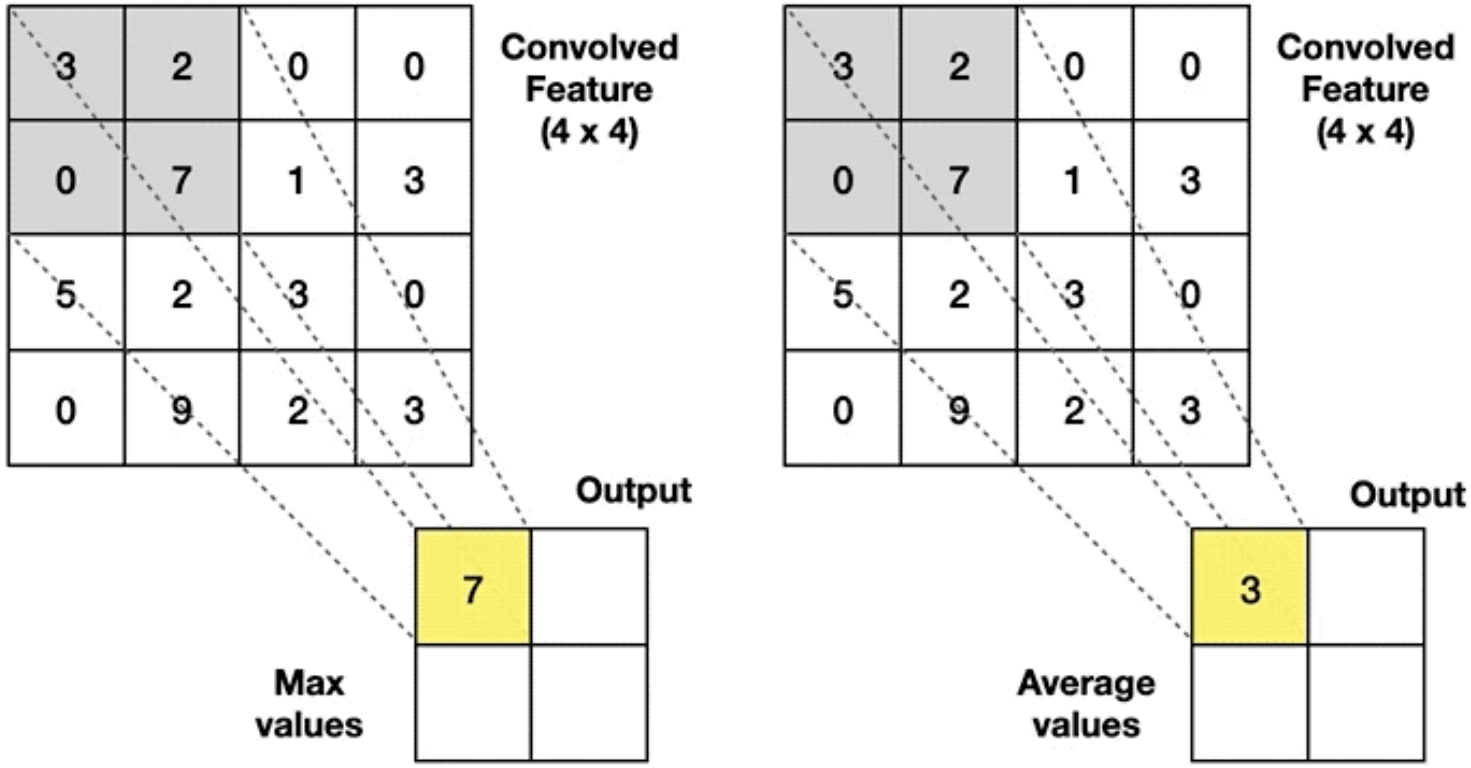


平均池化vs最大池化

1.2：池化层(Pooling Layer)

Max Pooling　　Average Pooling

```
y = MaxPooling2D(pool_size=(2, 2))(x)
y =Conv2DTranspose(32, 2, strides=(2, 2), padding='same',
activation='relu', use_bias=False, kernel_initializer='he_normal')(x)
```



实验8-1：

- 用Keras实现上面U-Net 结构，可视化重建结果、相对误差及Loss随Epoch变化。
- 对比下“去除与保留输出层的Skip Connection ”的效果，并分析原因。